



HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 3

CANDIDATE NAME

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CT GROUP

20S7 ____

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INDEX
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BIOLOGY

9816/01

Paper 1

20 Sept 2021

Insert

2 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

You may annotate this insert and use any blank pages for planning. **Do not write your answers** on the insert.

This document consists of **9** printed pages.

Information for Question 1

The information provided in this insert is based on a number of published scientific articles. Other published articles may not agree with all of this information.

Appetite is controlled by two systems

What, when and how much a person eats have typically been explained by two systems, one based on reward and one on hunger.

The reward system is more concerned with what type of food is eaten. At its heart is the dopamine pathway, which seems to respond most strongly to foods that are high in fat and sugar.

Fig 1.1 shows a diagram of a dopaminergic synapse, which is a chemical synapse where dopamine molecules are the neurotransmitter.

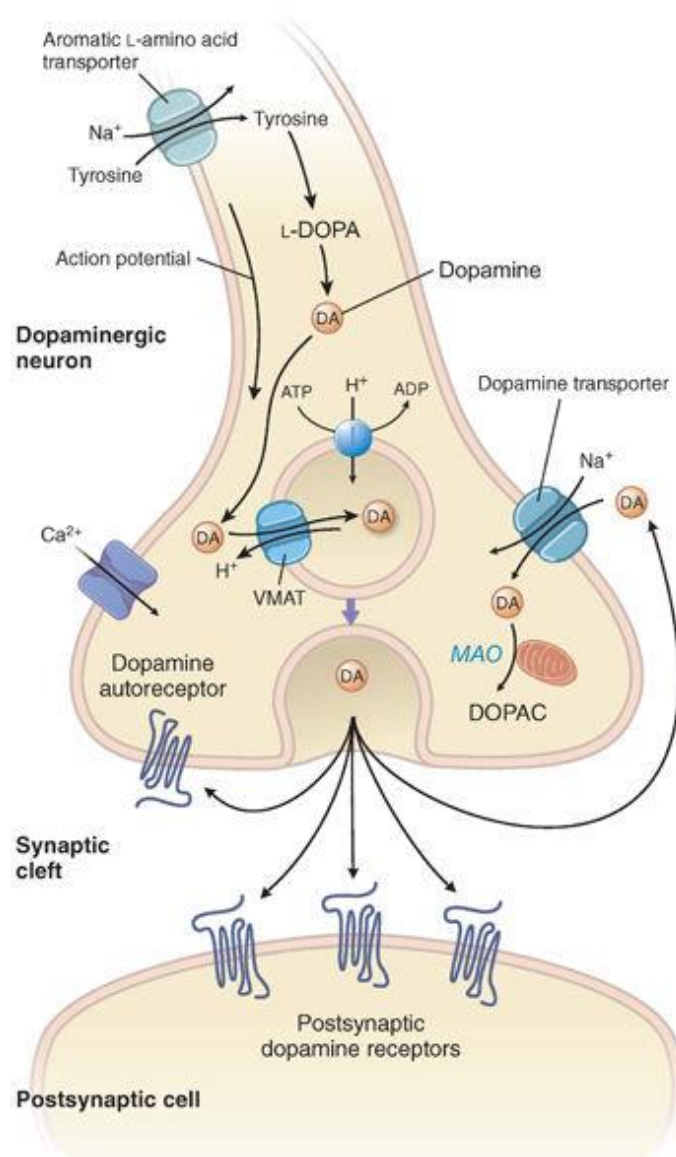


Fig. 1.1

The hunger system is mediated by hormones from the gut and from fat cells, which send information to the brain via the gut's own nervous system about when a person last ate and how hungry he or she should feel.

The stomach and small intestine produce numerous peptide hormones, including ghrelin. Fat cells release the peptide hormone leptin: the more fat the body has stored, the more leptin is released. Mutant mice that cannot produce leptin eat a lot, have low metabolic rates, and become obese.

Working together with a number of other hormones, ghrelin and leptin control appetite, metabolic rate and fat storage.

Ghrelin is released when the stomach is empty. Ghrelin stimulates hunger and eating, which tends to increase body weight and fat storage. Leptin is released when fat stores are high. Leptin decreases hunger and increases metabolic rate, which tends to decrease body weight and fat storage.

Fig 1.2 shows the roles played by hormones leptin and ghrelin in appetite regulation.

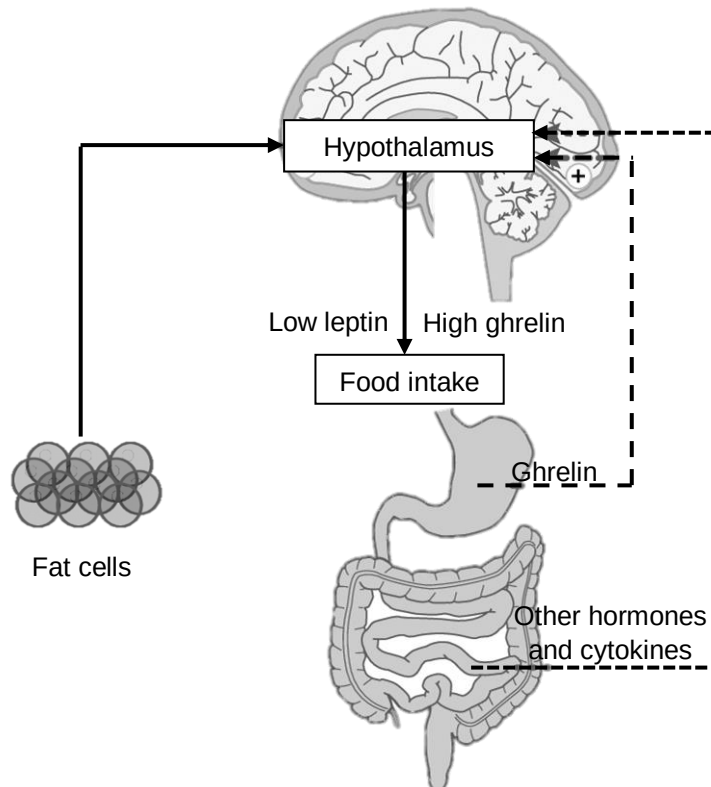


Fig. 1.2

Fig. 1.3 shows how the average plasma ghrelin (•) and leptin (○) concentrations change during a 24-hour period in 10 human subjects consuming breakfast (B), lunch (L), and dinner (D) at the times indicated.

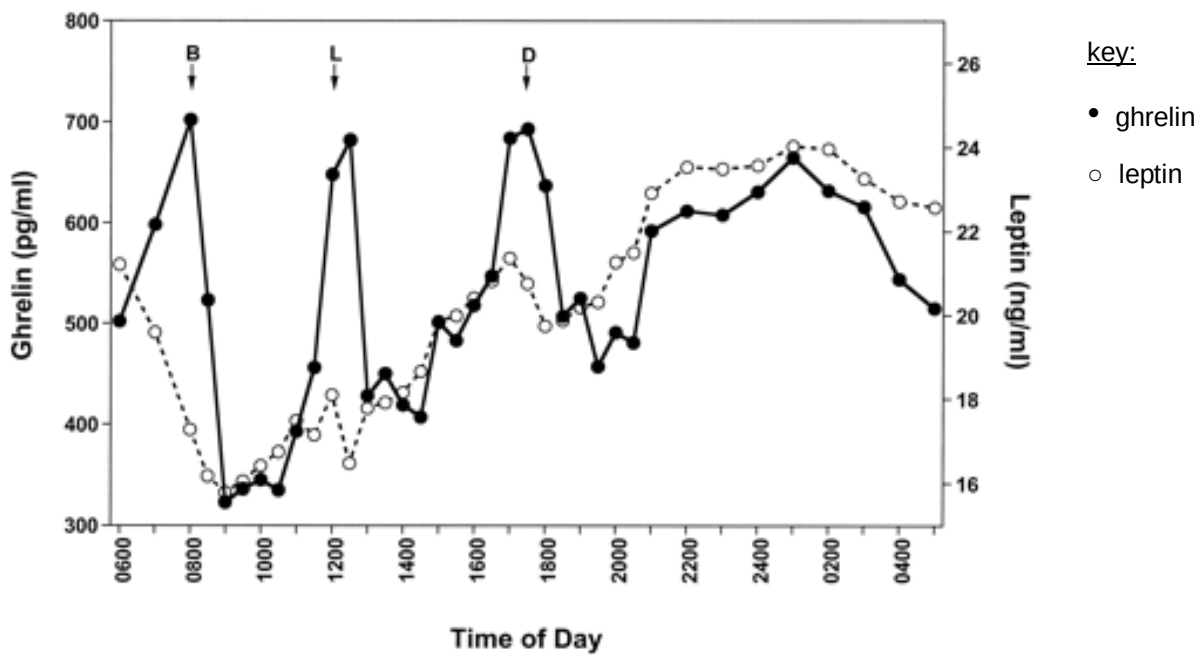


Fig. 1.3

The link between *FTO* gene and obesity

A gene called *FTO* is strongly linked to weight gain. There are two variants of the gene, a normal T allele and the mutant A allele.

A homozygous AA person's risk of becoming obese is increased by 70 percent as compared to a homozygous TT individual.

Fat mass and obesity-associated protein is an enzyme that in humans is encoded by the *FTO* gene located on chromosome 16.

Fig. 1.4(a) shows how the study was conducted:

- On the day prior to the study, 10 AA and 10 TT male human test subjects consumed the same evening meal at 20:00, then fasted and drank only water. On the study day, a venous cannula was inserted into a left forearm vein, and a period of 1 hour was allowed for acclimatization.
- At $t = 0$, a baseline blood sample and appetite measurement were taken. Subjects were then given a standard test meal to consume within 20 minutes.
- Blood was taken from the forearm at the timings indicated by the arrows, and concentration of circulating ghrelin present was measured and compared to the baseline.
- Subjective appetite was evaluated using visual analog scales (VAS) at these time intervals, and also compared to the baseline.

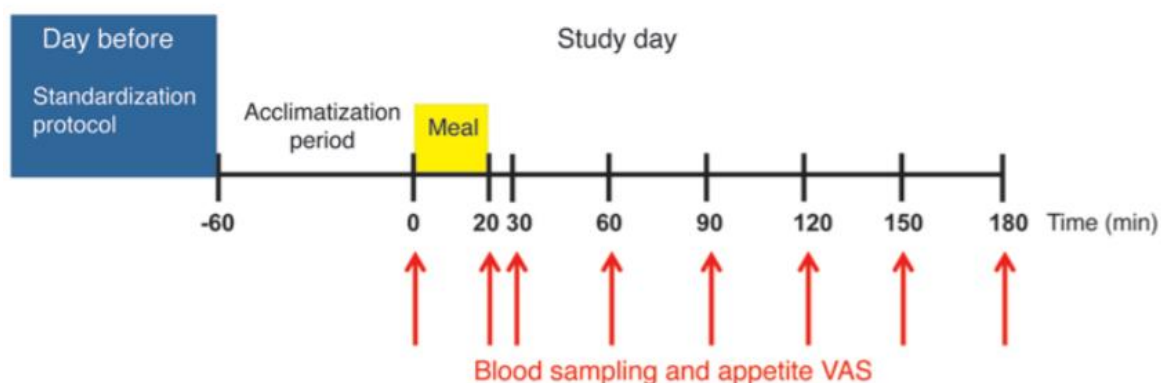


Fig. 1.4 (a)

Fig. 1.4(b) shows the results of the study.

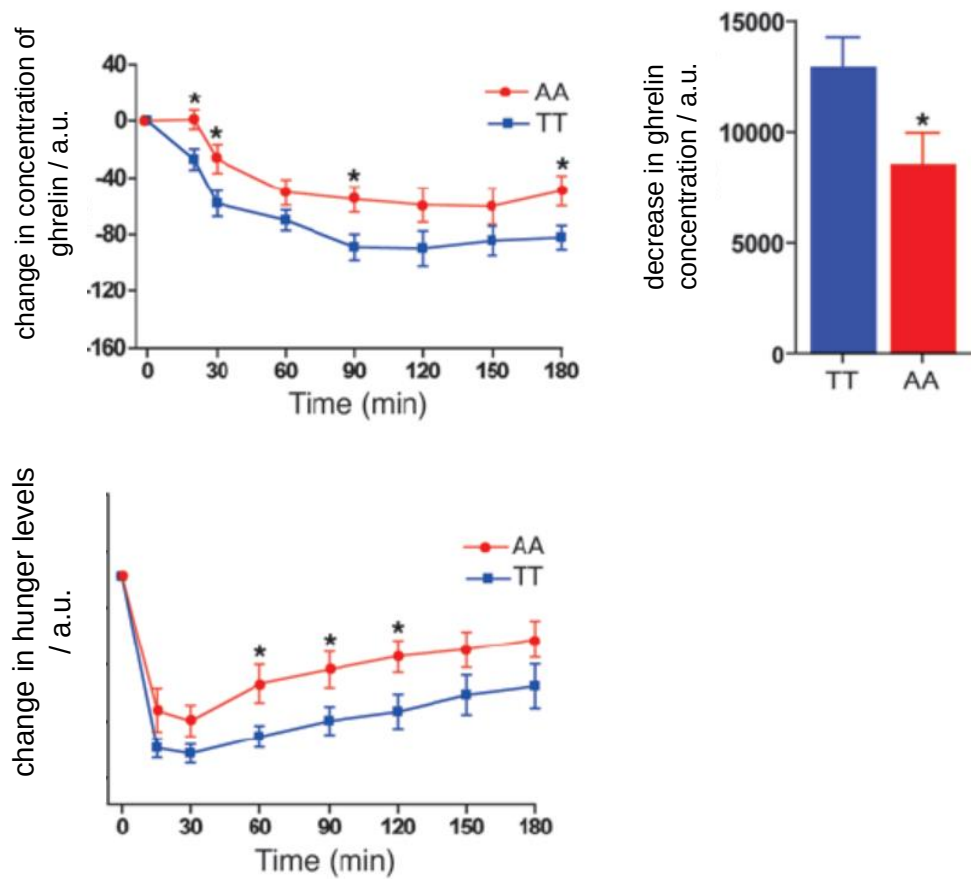


Fig. 1.4(b)

The scientists also made use of functional Magnetic Resonance Imaging (fMRI) to image the brain and measure the neural responses to visual food cues. The results are shown in Fig. 1.4 (c).

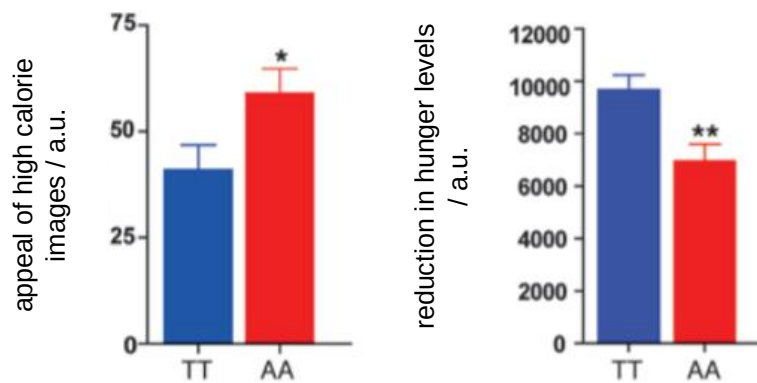


Fig. 1.4 (c)

A closer look into the functioning of FTO protein shows that it is involved in the demethylation of N₆-methyladenosine (m₆A) RNA, the most prevalent internal modification of messenger RNA (mRNA) in higher eukaryotes. Ghrelin mRNA has been identified as an FTO target.

Fig. 1.5 shows the effect of m₆A RNA methylation in cells.

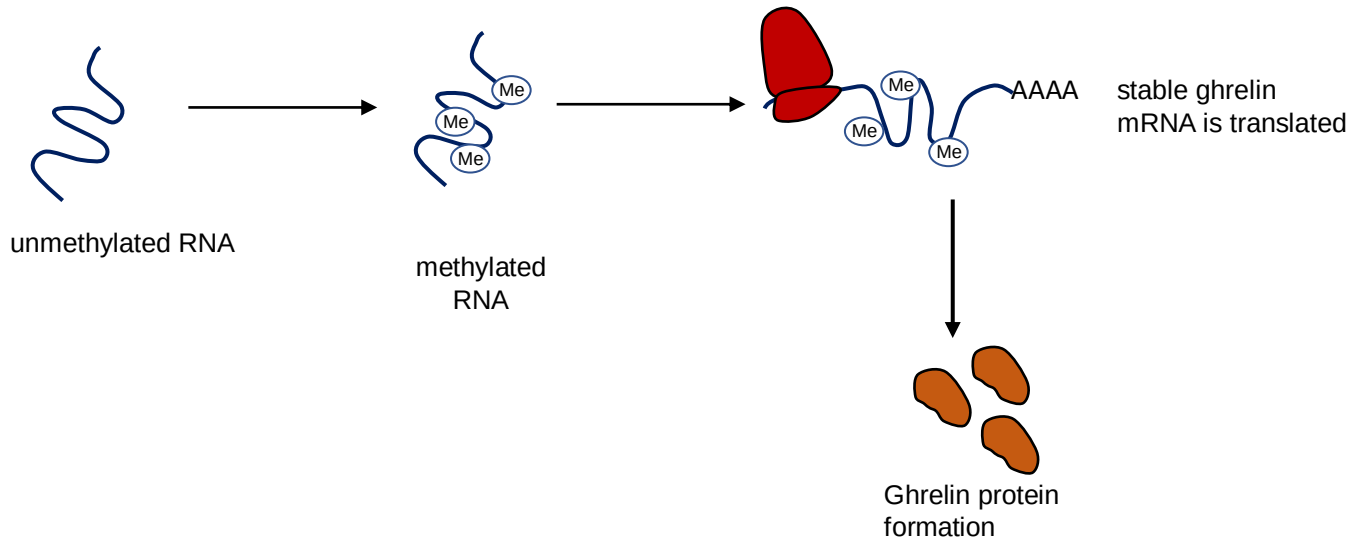


Fig. 1.5

Figure 1.6 shows all known data collected of FTO gene, presented in UniProt database.

The Fe2OG domain of the FTO protein spans from amino acids 32-327 and contains several mutations.

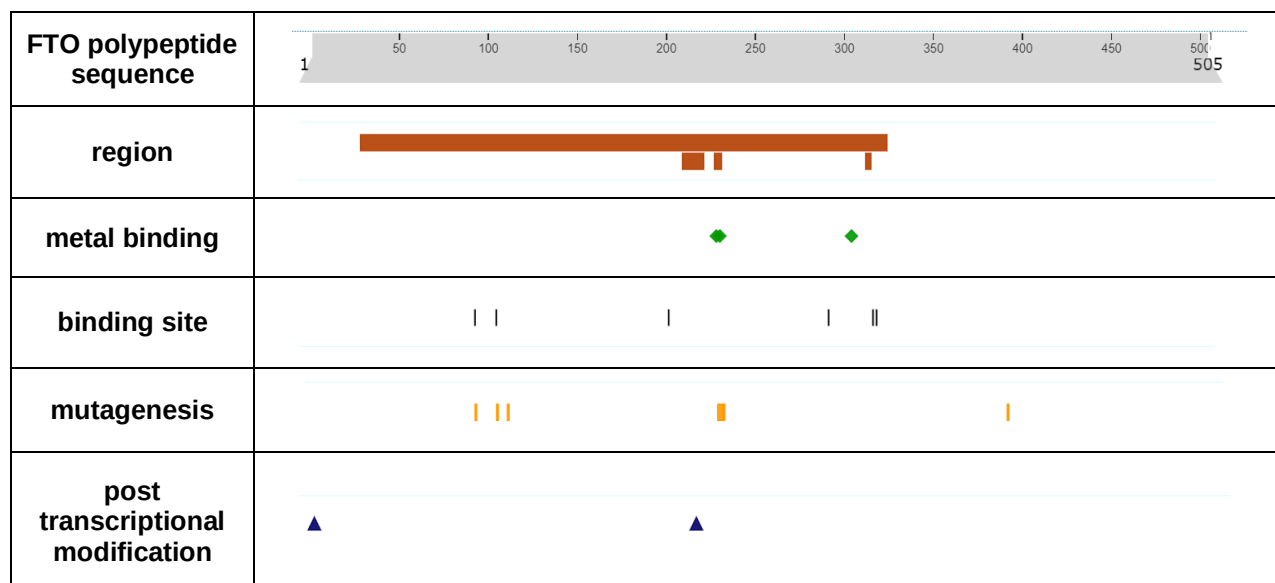


Fig. 1.6

Table 1.1 shows the various possible mutations that occur to the FTO enzyme and the position at which the mutagenesis occurs.

Table 1.1

position of mutagenesis	change in amino acid	description
114	phenylalanine ↓ aspartic acid	perturbs interaction between N-terminal and C-terminal domains and strongly reduces enzyme activity.
231-234	histidine-histidine-aspartic acid- glutamic acid ↓ alanine- histidine-alanine- proline	abolishes ability to demethylate N(6)-methyladenosine RNAs (m6A) RNAs.
392	cysteine ↓ aspartic acid	perturbs interaction between N-terminal and C-terminal domains and strongly reduces enzyme activity.

Modifying gut microbiota to treat obesity

The human body is host to a vast number of microbes, including bacterial, fungal and protozoal microorganisms, which together constitute our microbiota. Evidence is emerging that the intestinal microbiome is intrinsically linked with overall health, including obesity risk.

Obesity and obesity-related metabolic disorders are characterized by specific alterations in the composition and function of the human gut microbiome. Studies have shown that bacteria within the gut can influence energy utilisation from the diet and even influence host genes that regulate energy expenditure and storage.

This fact raises the attractive possibility that manipulating the gut microbiota could facilitate weight loss or prevent obesity in humans.

In 2014, Rice University synthetic biologist Jeff Tabor received a grant to genetically modify a harmless species of *Escherichia coli* bacteria normally found in the gut, in order to manipulate the intestinal environment to keep it healthy. He plans to engineer the cells to produce drugs when they and where they are called for – highly targeted antibiotics designed to bind with and deactivate bacterial chemicals that would otherwise leak into the blood from the intestine – thereby preventing the changes that lead to obesity, inflammation and associated ills.

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